# FLI 7-Mode to ONTAP transition workflow

ONTAP

NetApp
February 18, 2022

This PDF was generated from https://docs.netapp.com/us-en/ontap-fli/san-migration/concept_fli_7_mode_to_ontap_transition_workflow.html on February 18, 2022. Always check docs.netapp.com for the latest.

# Table of Contents

FLI 7-Mode to ONTAP transition workflow . . . 1
FLI 7-Mode to ONTAP transition workflow . . . 1
7-Mode to ONTAP FLI supported configurations . . . 1
Rebooting hosts . . . 1
Verify host LUN path and multipath configuration . . . 2
Prepare hosts for the transition . . . 2
Preparing source and destination arrays for migration . . . 2
Performing a disruptive cutover of FLI 7-Mode to ONTAP . . . 10
Importing the data from FLI 7-Mode to ONTAP . . . 12
Verifying FLI 7-Mode to ONTAP migration results . . . 12
FLI transition workflow post-migration tasks . . . 14

# FLI 7-Mode to ONTAP transition workflow

## FLI 7-Mode to ONTAP transition workflow

This section provides an example of the FLI 7-Mode to ONTAP transition workflow. The transition workflow can be run as either an online or offline workflow.

The FLI transition is recommended when either the source LUN is hosted on a 32-bit aggregate and/or the LUN is misaligned. The FLI 7-Mode to ONTAP transition is able to combine transitioning the LUN from 7-Mode to ONTAP, along with correcting LUN alignment and transitioning the LUN from a 32-bit to 64-bit aggregate. Other methods of transitioning the LUN, including 7-Mode Transition Tool (7MTT), can require remediating the LUN alignment and/or converting a 32-bit to 64-bit aggregate prior to transitioning to ONTAP.

The FLI 7-Mode to ONTAP transition workflow can be either an online or offline workflow. These workflows are functionally identical to the two corresponding FLI offline and online migration workflows, with the exception that the source array is a NetApp 7-Mode storage array. Both workflows share the same rules and procedures as their migration equivalents. This includes the FLI online workflow host operating support list.

The example provided should give a thorough walkthrough of the FLI 7-Mode to ONTAP process. The FLI 7-mode to ONTAP transition flow includes the following tasks:

1. Preparing source and destination arrays
2. Performing a disruptive cutover
3. Importing the data
4. Verifying migration results
5. FLI transition post-migration tasks

## 7-Mode to ONTAP FLI supported configurations

It is important to verify that the host OS, HBA, switch, and ONTAP array to which you are ultimately transitioning are supported.

If you are using the FLI 7-Mode to ONTAP transition workflow, you don’t need to verify your source (7-Mode controller) in the FlexArray IMT. It won’t be listed but is supported expressly for this transition workflow. You do still need to verify all hosts are in a supported configuration.

There are no FLI-specific platform requirements. There are also no minimum versions of 7-Mode Data ONTAP, although the version would have to support Fibre Channel protocol (FCP).

The maximum size LUN that FLI can import is 6 TB. This is a limitation based on the current maximum size drives currently supported by ONTAP. If you attempt to mount a larger foreign LUN, the LUN will be marked as broken, and you will not be able to write a label to it.

## Rebooting hosts

You have the option to reboot hosts prior to starting this workflow in order to verify that the host is in a known good state.

This would also be a good time to take a Snapshot copy in order to facilitate a revert if needed later. To verify

that the server configuration is persistent and pristine across reboots, complete the following steps:

**Steps**

1. Shut down all your open applications.
2. Review the logs for errors.
3. Verify that the host sees all of its paths.
4. Reboot the host.

# Verify host LUN path and multipath configuration

Prior to any migrations, verify that multipathing is correctly configured and working properly.

All available paths to LUNs should be active. See the SAN host multipath verification topics for examples of how to verify multipathing on Windows, Linux, and ESXi hosts.

# Prepare hosts for the transition

The execution phase includes the preparation of migration hosts.

In many instances it may be possible to have performed the remediation prior to this step. If not, then this is where you would perform any host remediation such as installing host attach kits or DSMs. From the analysis phase, you will have a gap list of items that need to be performed on each host in order for that host to be in a supported configuration using NetApp ONTAP. Depending on the type of migration being performed, either the host would be remediated and then rebooted (FLI 7-Mode to ONTAP online) or hosts would be rebooted, remediated, and then shut down (FLI 7-Mode to ONTAP offline).

# Preparing source and destination arrays for migration

To prepare for the FLI 7-mode to ONTAP migration, verify the host and source LUN paths and other details.

**Steps**

1. In ONTAP, change to `advanced` privilege level.

```
cluster::> set adv

Warning: These advanced commands are potentially dangerous; use them
only when directed to do so by NetApp personnel.
Do you want to continue? {y|n}: y

cluster::*>
```

2. Verify that the source array can be seen on the destination controller.

```
cluster::*> storage array show
Prefix                          Name   Vendor            Model Options
-------- ---------------------------- -------- ----------------
----------
NET-1                     NETAPP_LUN_1   NETAPP              LUN

cluster::*> storage array config show -array-name NETAPP_LUN_1
             LUN   LUN
Node         Group Count                    Array Name       Array Target
Port Initiator
------------ ----- ----- ----------------------------
----------------------- ---------
ontaptme-fc-cluster-01
                 1     2                 NETAPP_LUN_1
500a0981880b813d        0d

500a0981980b813d        0d
ontaptme-fc-cluster-02
                 1     2                 NETAPP_LUN_1
500a0981880b813d        0d

500a0981980b813d        0d
4 entries were displayed.

Warning: Configuration errors were detected.  Use 'storage errors show'
for detailed information.
```

3. View details on any storage errors listed. Some errors might require action before proceeding. However, the errors shown in the following example, “This device is an ONTAP® LUN.” can be safely ignored.

> The error message “This device is an ONTAP\(R\) LUN.” is caused because FLI relies on FlexArray technology which doesn’t support virtualizing ONTAP or ONTAP targets. FLI is able to import the LUNs, however, FlexArray doesn’t support virtualizing them.

```
cluster::*> storage errors show
Disk: NET-1.1
UID:
60A98000:44306931:452B4738:5767366B:00000000:00000000:00000000:00000000:
00000000:00000000
--------------------
NET-1.1 (60a9800044306931452b47385767366b): This device is an ONTAP(R)
LUN.

Disk: NET-1.2
UID:
60A98000:44306931:452B4738:5767366D:00000000:00000000:00000000:00000000:
00000000:00000000
--------------------
NET-1.2 (60a9800044306931452b47385767366d): This device is an ONTAP(R)
LUN.

2 entries were displayed.
```

4. Display source LUN details.

```
cluster::*> storage array config show -array-name NETAPP_LUN_1 -instance

           Controller Name: ontaptme-fc-cluster-01
                 LUN Group: 1
        Array Target Ports: 500a0981880b813d
                 Initiator: 0d
                Array Name: NETAPP_LUN_1
   Target Side Switch Port: stme-5010-4:2-6
Initiator Side Switch Port: stme-5010-4:2-3
      Number of array LUNs: 2

           Controller Name: ontaptme-fc-cluster-01
                 LUN Group: 1
        Array Target Ports: 500a0981980b813d
                 Initiator: 0d
                Array Name: NETAPP_LUN_1
   Target Side Switch Port: stme-5010-4:2-5
Initiator Side Switch Port: stme-5010-4:2-3
      Number of array LUNs: 2

~~~~~~ Output truncated ~~~~~~~
4 entries were displayed.

Warning: Configuration errors were detected.  Use 'storage errors show'
for detailed information.
```

5. Verify the source array is discovered through all the initiator ports.

```
cluster::*> storage array config show -array-name NETAPP_LUN_1
             LUN   LUN
Node         Group Count                   Array Name       Array Target
Port Initiator
------------ ----- ----- ----------------------------
----------------------- ---------
ontaptme-fc-cluster-01
                 1     2                 NETAPP_LUN_1
500a0981880b813d        0d

500a0981980b813d        0d
ontaptme-fc-cluster-02
                 1     2                 NETAPP_LUN_1
500a0981880b813d        0d

500a0981980b813d        0d
4 entries were displayed.

Warning: Configuration errors were detected.  Use 'storage errors show'
for detailed information.
```

6. List the LUNs mapped from the 7-Mode storage. Verify the disk properties and paths.

```
cluster::*> storage disk show -array-name NETAPP_LUN_1 -instance
                  Disk: NET-1.1
        Container Type: unassigned
            Owner/Home: -  / -
               DR Home: -
    Stack ID/Shelf/Bay: -  / -  / -
                   LUN: 0
                 Array: NETAPP_LUN_1
                Vendor: NETAPP
                 Model: LUN
         Serial Number: D0i1E+G8Wg6k
                   UID:
60A98000:44306931:452B4738:5767366B:00000000:00000000:00000000:00000000:
00000000:00000000
                   BPS: 512
         Physical Size: -
              Position: present
Checksum Compatibility: block
             Aggregate: -
                  Plex: -
Paths:
                                LUN  Initiator Side        Target Side
```

```
Link
Controller         Initiator     ID  Switch Port           Switch Port
Acc Use  Target Port                TPGN    Speed      I/O KB/s
IOPS
------------------ ---------  -----  --------------------
--------------------  --- ---  -----------------------  ------  -------
------------  ------------
ontaptme-fc-cluster-02
                   0d             0  stme-5010-4:2-4       stme-5010-
4:2-6       ANO RDY  500a0981880b813d              1   4 Gb/S
0             0
ontaptme-fc-cluster-02
                   0d             0  stme-5010-4:2-4       stme-5010-
4:2-5       AO  INU  500a0981980b813d              0   4 Gb/S
0             0
ontaptme-fc-cluster-01
                   0d             0  stme-5010-4:2-3       stme-5010-
4:2-6       ANO RDY  500a0981880b813d              1   4 Gb/S
0             0
ontaptme-fc-cluster-01
                   0d             0  stme-5010-4:2-3       stme-5010-
4:2-5       AO  INU  500a0981980b813d              0   4 Gb/S
0             0

Errors:
NET-1.1 (60a9800044306931452b47385767366b): This device is a ONTAP(R)
LUN.
~~~~~~ Output truncated ~~~~~~~
2 entries were displayed.
```

7. Verify the source LUN is marked as foreign.

```
cluster::*> storage disk show -array-name NETAPP_LUN_1
                     Usable           Disk    Container   Container
Disk                   Size Shelf Bay Type    Type        Name
Owner
---------------- ---------- ----- --- ------- ----------- ---------
--------
NET-1.1                   -     -   - LUN     unassigned  -         -
NET-1.2                   -     -   - LUN     foreign     -         -
2 entries were displayed.
```

8. Serial numbers are used in FLI LUN import commands. List all foreign LUNs and their serial numbers.

```
cluster::*> storage disk show -container-type foreign -fields serial-
number
disk    serial-number
------- ------------------------------
NET-1.2 D0i1E+G8Wg6m
```

9. Create the target LUN. The `LUN create` command detects the size and alignment based on partition offset and creates the LUN accordingly with the foreign-disk argument

```
cluster::*> vol create -vserver fli_72C -volume flivol -aggregate aggr1
-size 10G
[Job 12523] Job succeeded: Successful
```

10. Verify volume.

```
cluster::*> vol show -vserver fli_72C
Vserver   Volume       Aggregate    State      Type       Size
Available Used%
--------- ------------ ------------ ---------- ---- ----------
---------- -----
fli_72C   flivol       aggr1        online     RW         10GB
9.50GB    5%
fli_72C   rootvol      aggr1        online     RW          1GB
972.6MB    5%
2 entries were displayed.
```

11. Create the target LUN.

```
cluster::*> lun create -vserver fli_72C -path /vol/flivol/72Clun1
-ostype windows_2008 -foreign-disk D0i1E+G8Wg6m

Created a LUN of size 3g (3224309760)
```

12. Verify new LUN.

```
cluster::*> lun show -vserver fli_72C
Vserver   Path                            State   Mapped   Type
Size
--------- ------------------------------- ------- -------- --------
--------
fli_72C   /vol/flivol/72Clun1             online  unmapped windows_2008

3.00GB
```

13. Create an igroup of protocol FCP with host initiators.

```
cluster::*> lun igroup create -vserver fli_72C -igroup 72C_g1 -protocol
fcp -ostype windows –initiator 10:00:00:00:c9:e6:e2:79

cluster::*> lun igroup show -vserver fli_72C -igroup 72C_g1
          Vserver Name: fli_72C
           Igroup Name: 72C_g1
              Protocol: fcp
               OS Type: windows
Portset Binding Igroup: -
           Igroup UUID: 7bc184b1-dcac-11e4-9a88-00a0981cc318
                  ALUA: true
            Initiators: 10:00:00:00:c9:e6:e2:79 (logged in)
```

14. Map the test LUN to the test igroup.

```
cluster::*> lun map -vserver fli_72C -path /vol/flivol/72Clun1 -igroup
72C_g1

cluster::*> lun mapping show -vserver fli_72C
Vserver    Path                                      Igroup   LUN ID
Protocol
---------- ----------------------------------------  -------  ------
--------
fli_72C    /vol/flivol/72Clun1                       72C_g1        0
fcp
```

15. Offline the test LUN.

```
cluster::*> lun offline -vserver fli_72C -path /vol/flivol/72Clun1

Warning: This command will take LUN "/vol/flivol/72Clun1" in Vserver
"fli_72C" offline.
Do you want to continue? {y|n}: y

cluster::*> lun show -vserver fli_72C
Vserver   Path                            State   Mapped   Type
Size
--------- ------------------------------- ------- -------- --------
--------
fli_72C   /vol/flivol/72Clun1             offline mapped   windows_2008

3.00GB
```

16. Create import relationship between new LUN and foreign LUN.

```
cluster::*> lun import create -vserver fli_72C -path /vol/flivol/72Clun1
-foreign-disk D0i1E+G8Wg6m

cluster::*> lun import show -vserver fli_72C -path /vol/flivol/72Clun1
vserver foreign-disk   path                operation admin operational
percent
                                          in progress state state
complete
---------------------------------------------------------------------------
-------
fli_72C D0i1E+G8Wg6m   /vol/flivol/72Clun1 import    stopped
                                                            stopped
0
```

# Performing a disruptive cutover of FLI 7-Mode to ONTAP

This example gives you the general steps for performing a disruptive cutover for the FLI transition process.

For a host remediation walkthrough of Windows, Linux, and ESXi hosts, see related topics in this guide as well as the host operating system and host-attach kit documentation.

**Steps**

1. On the 7-Mode system, display the igroup to which the source LUN is mapped.

```
stme-7ma> igroup show
    FLI_on_fcp (FCP) (ostype: windows):
        10:00:00:00:c9:e6:e2:79 (logged in on: 0c, vtic)
        50:0a:09:81:00:96:43:70 (logged in on: 0c, vtic)
        50:0a:09:81:00:96:3c:f0 (logged in on: 0c, vtic)
```

> The disruption begins immediately after the unmap command is executed. Generally, the disruption window can be measured in minutes. It is literally the length of time it takes to move the host to the new NetApp target and scan for LUNs.

2. If the LUNs being imported are for ESXi hosts, review and follow the instructions in the *ESXi CAW/ATS remediation* topic.
3. Use the `unmap` command to move the LUN from its hosts. (The disruption window begins here.)

```
stme-7ma> igroup remove -f FLI_on_fcp 10:00:00:00:c9:e6:e2:79
```

4. Verify that the host initiators are no longer present.

```
stme-7ma> igroup show
    FLI_on_fcp (FCP) (ostype: windows):
        50:0a:09:81:00:96:43:70 (logged in on: 0c, vtic)
        50:0a:09:81:00:96:3c:f0 (logged in on: 0c, vtic)
```

5. On the ONTAP cluster, bring the destination LUN online and verify that it's mapped.

```
cluster::*> lun online -vserver fli_72C -path /vol/flivol/72Clun1

cluster::*> lun show -path /vol/flivol/72Clun1
Vserver   Path                            State   Mapped   Type
Size
--------- ------------------------------- ------- -------- --------
--------
fli_72C   /vol/flivol/72Clun1             online  mapped   windows_2008

3.00GB
```

6. Rescan disks on the host; find the LUN on the ONTAP target.

> The disruption window ends here.

The LUNs are online and mapped, and hosts are now mounting the new ONTAP hosted LUN. Reads are passed through the ONTAP array to the source LUN, and writes are written to both the new ONTAP hosted LUN and also the original source LUN. Both source and destination LUNs will remain in sync until the

migration is complete and the LUN relationship has been broken.

# Importing the data from FLI 7-Mode to ONTAP

These steps describe how to import the data from a 7-Mode source LUN to an ONTAP destination LUN using FLI.

**Steps**

1. Start the migration import.

```
cluster::*> lun import start -vserver fli_72C -path /vol/flivol/72Clun1
```

2. Display FLI status.

```
cluster::*> lun import show -vserver fli_72C -path /vol/flivol/72Clun1
vserver foreign-disk   path                operation admin operational
percent
                                         in progress state state
complete
---------------------------------------------------------------------------
-------
fli_72C D0i1E+G8Wg6m   /vol/flivol/72Clun1 import    started
                                                           completed
100
```

If you want to make sure the source LUN remains consistent after the migration completes, you will need to:

- After the import show indicates it is completed, shut down the host.
- Delete the LUN relationship: `lun import delete -vserver fli_72C -path /vol/flivol/72Clun1`.

> Remember that after the LUN relationship is broken, the LUNs will quickly lose sync because changes are made to the new LUN only. Therefore, while it may be beneficial to maintain a consistent state in the event you want to restore the original state, the new LUN will likely have changes not reflected in the source LUN.

> After the import is stopped, you can destroy the import relationship unless you intend to verify the import.

# Verifying FLI 7-Mode to ONTAP migration results

You have the option to verify that LUNs have been properly migrated from FLI 7-Mode to ONTAP.

Start the verify job to compare source and destination LUNs. Monitor the verify progress. The LUNs being verified need to be offline for the duration of the verification session. The verification session can potentially be lengthy because it is a block-for-block comparison between source and destination LUNs. It should take approximately the same amount of time as the migration. Verification is not required, but we encourage you to verify a subset of the LUNs imported/migrated to feel comfortable about the import process.

> The LUN import verify must be explicitly stopped before bringing the LUN back online. Otherwise, the LUN online fails. This behavior will be changed in an upcoming release of ONTAP.

**Steps**

1. Offline the LUNs to be verified.

```
cluster::*> lun offline -vserver fli_72C -path /vol/flivol/72Clun1
Warning: This command will take LUN "/vol/flivol/72Clun1" in Vserver
"fli_72C" offline.
Do you want to continue? {y|n}: y
```

2. Start LUN verify.

```
lun import verify start -vserver fli_72C -path /vol/flivol/72Clun1
```

3. Display LUN verify status.

```
ontaptme-fc-cluster::*> lun import show -vserver fli_72C -path
/vol/flivol/72Clun1
vserver foreign-disk   path                operation admin operational
percent
                                         in progress state state
complete
---------------------------------------------------------------------------
-------
fli_72C D0i1E+G8Wg6m   /vol/flivol/72Clun1 verify    started

9
```

> The LUN import verify must be explicitly stopped before bringing the LUN back online. Otherwise, the LUN online fails. See the following CLI output.

4. Stop LUN verification. This step needs to be performed manually even if the status shows that verify is complete.

```
lun import verify stop -vserver fli_72C -path /vol/flivol/72Clun1
```

5. Online the LUN upon completion of verification.

```
lun online -vserver fli_72C -path /vol/flivol/72Clun1
```

# FLI transition workflow post-migration tasks

The post-migration tasks for the FLI 7-mode to ONTAP workflow are similar to the other FLI workflows.

- When ready, you can delete the LUN import relationship.

  The LUN import relationship can be safely removed because the host is now accessing the new NetApp array for all I/O to the new ONTAP LUN, and the source 7-Mode LUN is no longer in use.

- All server remediations are performed during post-migration.

  The third-party software is removed, NetApp software is installed and configured, and then the host is brought up accessing the LUNs on NetApp.

- Review logs for errors, check pathing, and perform any application testing to verify that your migration completed cleanly and successfully.

**Copyright Information**

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

**Trademark Information**

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.